

Prevention of Gestational Failure by High Protein Diet

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A high protein, low carbohydrate diet program has been in continuous use in a specific clinic population for twenty years. Results from 12,000 maternities show substantial reductions in the frequency of gestosis of all degrees of severity, abruptio placentae, inexplicable fetal death, and overall and corrected perinatal mortality. The results show a direct relation to the success of dietary management as monitored by an index which accounts for both weight gain and hemoglobin level. Half of the patients achieved a high diet success. In this group these several conditions occurred only at very low frequencies and both severe gestosis and abruptio placentae were for practical purposes eliminated. By contrast the group with the least success with the diet program had very high frequencies of all these same complications.

Additional key words: toxemia, hypertensive pregnancy, nutrition, perinatal death, section delivery, hemoglobin level.

The search for the cause and prevention of gestational failures, including so-called toxemia (gestosis) is not new. Further, the importance of diet, even high protein diet is not new. Beginning in 1938, when I first came to Motherwell Maternity, I set out to find means of preventing pre-eclampsic toxemia and abruptio placentae, and such problems as difficult forceps, section birth, and perinatal death. Various considerations led me to diet as central to these issues, as fundamental to the health of the pregnant woman and her baby. During gestation there is a stockpiling of protein in the form of a healthy, mature baby, an efficient uterus and birth canal, growth of the breasts, and expansion of both the vascular bed and the circulating blood volume. To meet these needs I have developed a high protein diet scheme and have used the blood hemoglobin level against the weight gain as a means of monitoring the progress of the patient. My basic diet program has been in use since 1953 and the overall comprehensive management program since 1959.

The Motherwell area

The Burgh of Motherwell and Wishaw, 14 miles from

Glasgow, is a steel manufacturing town of 75,000 persons, with an average annual birth registration of 1,400 during 1960-1971. The average annual perinatal death rate was 27 per 1000 and the lowest annual rate during this same interval was 19 per 1000.

The basic diet in western Scotland is poor. It contains a great excess of starchy food. The incidence of high risk maternities (teen-age primigravidas, elderly primigravidas, and especially gravida 5 and over) is 13% which is above the average for Scotland. These are very ordinary people, the wives of factory workers, shopkeepers, clerks, drivers, steel workers, and general laborers. The low socio-economic groups are strongly represented.

The hospital and program

Motherwell Maternity Hospital has 37 beds, 1 resident house officer, and 1 part time consultant. Annual admissions run to 1,500 per year including abortions, and viable births average 1,000 per year, 92% booked. The hospital, accordingly, accounts for over 70% of the births in the Burgh each year. The staff includes midwives (maternity nurses), nurses, and trained cooks. We have had a training school for midwives for 50 years. Two clinics are part of the program, one in Motherwell and one in Wishaw. General practitioners may participate in the antenatal care of their own patients.

The gestational index

The index, first formalized in 1956, is derived according to the clinical findings in the 38th week (Table I). Group 2B is the pivotal group, with some results better than the low group (3B, 3W and 3BW), but others about the same. For convenience groups 1 and 2A are designated as *high* protein success, group 2B as *medium*, groups 3B and 3W as *low*, and group 3BW as *very low*. Weight control (W) is not an end in itself, but as a means to facilitate protein intake, to correct the excesses of starch in the usual regional diet, and to stabilize blood volumes for validation of the hemoglobin level as the index of protein intake (B).

The diet

As described in handouts and lectures for the women, the diet includes high protein intake and low starch intake. It is a special diet. Changes in it reduce its special

TABLE I

GESTATIONAL INDEX SCHEME BY DIET GRADES AND SUCCESS GROUPS

HEMOGLOBIN LEVEL AS g/dl	WEIGHT GAIN PER WEEK AVERAGE			GROUP
	<250g	250-500g	500+g	
12+	1	2A	-	LOW GROUP
10-12	2A	2B	3W	
<10	-	3B	3BW	
				VERY LOW GROUP

Iron and folic acid supplements are not prescribed except in case of dietary failure when the hemoglobin level is approaching 10 g/dl. Diuretics are not used to control weight gain. A two day course may be used but only in hospital to demonstrate excess fluid retention. Salt restriction is harmful and is not part of the regimen.

quality. There are twelve points. Diuretics and salt restriction are not used.

- (a) *Meat.* One pound of red meat should be eaten every day of gestation. Quantity is more important than quality. It will be necessary to eat it between meals as well as at meals and cold meat, especially corned beef or tongue is good for the purpose. Liver, white meat, sausages, black pudding (blood sausage) and meat roll are not so good, but can be eaten in small quantities to provide variety.

Sometimes nausea or even vomiting is apparently caused by eating meat. Please do not stop eating it on this account but report the complaint and keep on eating.

When hungry between meals eat MEAT.

- (b) *Fish, eggs and cheese.* Valuable sources of body building food, but should not be eaten in large quantities.
- (c) *Soup.* Thin or strained soup is allowed. Do not take broth, cream, noodle, lentil or pea soup.
- (d) *Vegetables.* Try to eat vegetables twice daily. These are cabbage, sprouts, cauliflower, spinach,

celery, parsley, leeks, lettuce, tomatoes, cucumber, marrow and mushrooms. Do not eat peas, beans, turnips, parsnips, carrots or beetroot.

- (e) *Fruit.* The best fruits are oranges and grapefruits. Apples, pears, grapes should be limited. Fruit should be eaten as a course and not between meals. Do not eat prunes, plums, bananas, and tinned fruits or nuts and dates.
- (f) *Drinks.* Tea, cocoa, coffee made with water and with a little milk added are best. Sugar may be added, but not more than four heaped teaspoonsfuls per day. When thirsty avoid excessive drinking. Do not use more than 1/2 pint milk per day, for all purposes. Do not drink aerated (carbonated) waters.
- (g) *Sweets.* Sweets should be limited to 1/2 pound of boiled sweets (hard candy) per week. Do not eat chocolate.
- (h) *Do not eat potatoes or chips.* Bread — either white, brown, malted or toasted, rolls, scones, cakes, or biscuits of any kind. (Includes water biscuits, cream crackers and digestive, chocolate.)
- (i) *Do not eat patent foods* such as are claimed by the manufacturers to be low calorie, weight-limiting products.
- (j) *Do not eat milk puddings* (sago, rice, semolina, tapioca and others). Cereals (porridge, cornflakes, crispies and others), macaroni, spaghetti and ice cream.
- (k) *Smoking* is bad and expensive. It is a recognized cause of premature birth. It interferes with digestion. Smokers must at least reduce to one cigarette after each meal. But better to stop altogether.
- (l) *If you persevere* with this diet for three weeks it becomes natural and easy. It must be admitted that the change over is difficult.

The distribution between diet groups by age and parity

Index grades 1 and 2A, the high group, include a good proportion of high risk maternities (Table II). In 12,027 consecutive cases there were 286 elderly primigravidae out of 5,808 women in grades 1 + 2A (4.9%). Grand multigravidae, 5 or more pregnancies, accounted for 9.8%. In all, almost half achieved high protein intake. Elderly primigravidae were above half, teenage primigravidae and grand multigravidae below. All parities and age groups have sufficient representatives for meaningful analysis, and the range of success in the diet progress permits comparison across that range.

TABLE II
DISTRIBUTION OF PATIENTS ACCORDING TO AGE AND GRAVIDITY
WITH PROPORTION ACHIEVING HIGH DIETARY SUCCESS

	< 19 y	Primigravidae 20-29 y	30 + y	Multigravidae Gravidae 2-4	Gravidae 5 +
Achieved grade					
1 or 2A (high)	323	1738	286	2894	567
2B, 3W, 3B and 3 BW (medium, low and very low)	658	1434	118	2999	959
Totals	981	3172	404	5893	1526
Percent as grade 1 or 2A	33	55	71	49	37

Based on 12027 consecutive cases booked and unbooked during 1960-1971, with 51 cases having incomplete data. Total primigravidae with complete data = 4557; multigravidae = 7419. 2347 primigravidae achieved high diet success (51.5%); 3461 multigravidae the same (46.7%).

RESULTS

Gestosis

Patients were classified as one of three grades of gestosis (so-called toxemia). These were defined as:

1. Gestosis E: all women with ankle edema without hypertension or proteinuria.

2. Gestosis EH: all women with or without ankle edema plus blood pressure over 120/90 Torr.

3. Gestosis EPH: all three signs present, ankle edema, hypertension, proteinuria. For plotting on Figure 1 and for listing in Table III, four cases of convulsive gestosis are listed with EPH gestosis. These cases were in the low and very low diet success groups.

The table and the graph show clearly that the risk is related to diet success. High hemoglobin and reasonable weight gain mean a very low risk of gestosis (5.5 per 1000). The risk for the very low success group is 35 times higher.

Other complications

Patients were classified as having essential hypertension only when elevated diastolic blood pressure persisted at the six weeks postpartum examination. Instances of essential hypertension + gestosis are classed as gestosis. Table IV shows that the rate of essential hypertension is more or less the same in the four diet groups.

By contrast, a further inverse relation between diet success and abruptio placentae is shown. The risk for the very low diet success group is about 160 times greater than the composite risk for the medium and high success groups.

Although there is also a wide range of risk of inexplicable fetal death, there is no difference in these figures between the medium and low success diet groups. It is possible that inexplicable fetal death is the consequence of these nutritional factors acting in some conjoint way rather than by additive means.

Perinatal death rate

Certain anomalies with a poor prognosis are common in Western Scotland. Included are anencephaly, hydrocephaly, severe spina bifida with myelomeningocele, and congenital heart disease. The frequency of these is essentially constant as among the various diet groups, but

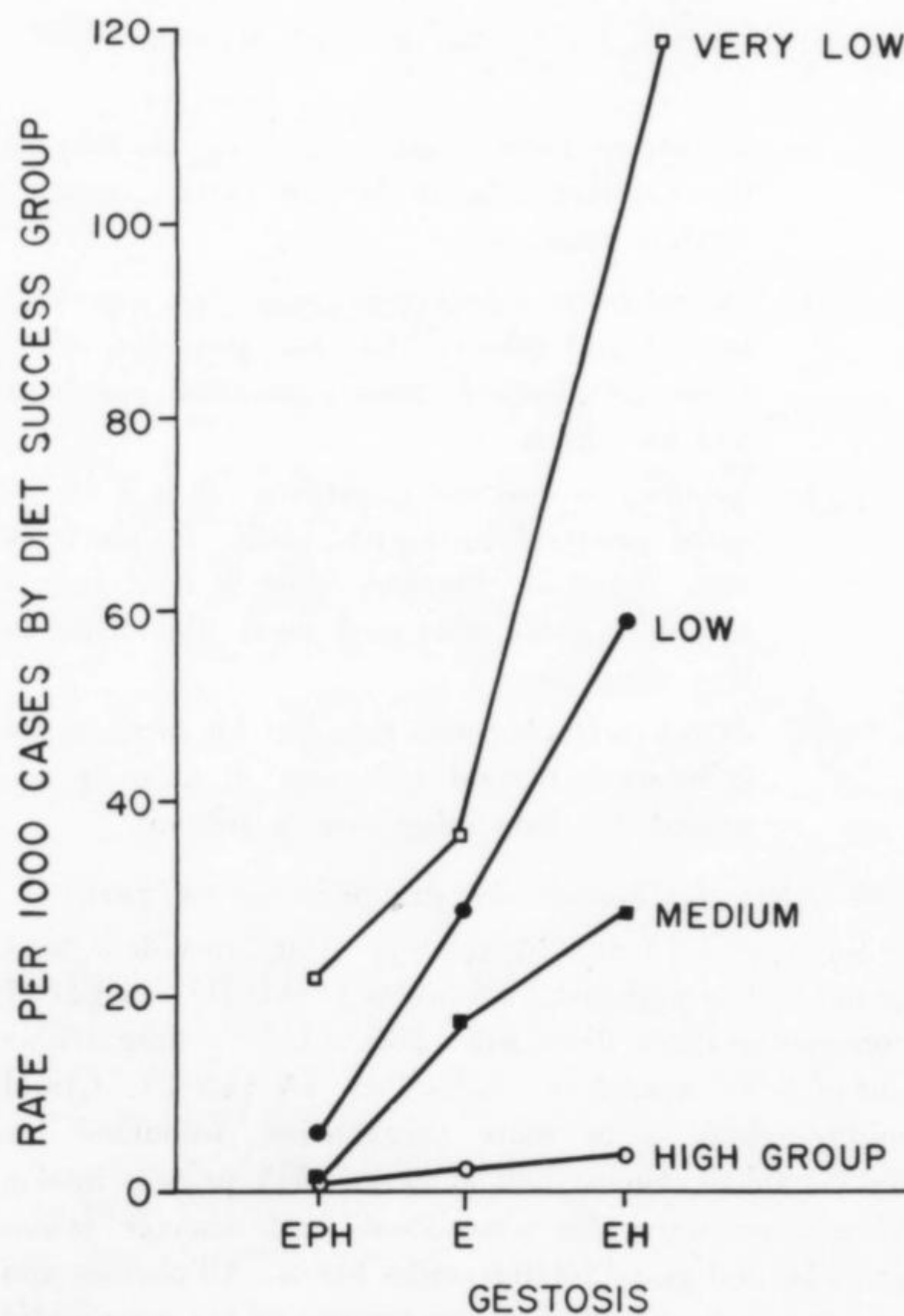


FIG. 1. Frequency of gestosis according to degree of diet success. EPH = edema, proteinuria, and hypertension; E = edema only; EH = edema and hypertension.

premature birth (before the 37th week) does show a variation over the range of diet success. Table V gives the gross perinatal death rates and the corrections due to deformity incompatible with survival and to premature birth. The corrected result again supports the value of high protein, low carbohydrate diet during pregnancy.

Surgically assisted delivery

In the interval 1960-1971 there were 99 section deliveries out of 12,027 maternities, a rate of 0.82%. There were 8,000 planned inductions with only 20 sections

performed, a rate of 0.25%. This leaves a residual frequency of 79 sections/4,027 non-induced labors, or 1.96%. Forceps intervention was also infrequent, with low forceps used in 5.4% and middle forceps in 0.9% of the 8,000 induced cases.

COMMENT

After many years of observation of the relationship between high hemoglobin and the success of pregnancy on the one hand, and the relationship between diet and hemoglobin on the other, I chose a particular diet program for

TABLE III
FREQUENCY OF GESTOSIS BY DIET SUCCESS GROUPING

Degree of gestosis	High group (Grades 1 + 2A)		Medium group (Grade 2B)		Low group (Grades 3B + 3W)		Very low group (Grade 3BW)	
	No.	Per 1000	No.	Per 1000	No.	Per 1000	No.	Per 1000
Gestosis E only	14	(2.4)	28	(18.4)	113	(28.9)	27	(36.7)
Gestosis EH	19	(3.2)	42	(27.6)	226	(57.8)	88	(119.7)
Gestosis EPH	1	(0.2)	0	(0.0)	22*	(5.6)	16*	(21.8)
All forms of gestosis	34	(5.8)	70	(46.0)	361*	(92.3)	131*	(178.2)
Total number per diet group	5,808		1,523		3,910		735	

Based on 12,027 cases during the interval 1960-1971 with 51 cases incomplete. The total number of cases of gestosis E = 182 (1.5%); of gestosis EH = 375 (3.1%), and of EPH = 39 (0.32%).*

*Includes four cases of convulsive gestosis or eclampsia.

TABLE IV
FREQUENCY OF VARIOUS COMPLICATIONS IN 11,976 MATERNITIES
ACCORDING TO DIET SUCCESS GROUPING*

Complication	High group (Grades 1 + 2A)		Medium group (Grade 2B)		Low group (Grades 3B + 3W)		Very low group (Grade 3BW)	
	No.	Per 1000	No.	Per 1000	No.	Per 1000	No.	Per 1000
Abruptio placentae	1	(0.2)	1	(0.6)	32	(8.2)	32	(43.5)
Inexplicable fetal death	3	(0.5)	9	(5.9)	20	(5.1)	16	(21.8)
Essential hypertension	221	(38.1)	40	(26.3)	112	(28.6)	23	(31.3)

*During the period 1960-1971: the totals in each diet group are the same as in Table III.

TABLE V
PERINATAL DEATH RATES BY DIET GROUPS IN 17,976 MATERNITIES*

	High group (Grades 1 + 2A)		Medium group (Grade 2B)		Low group (Grades 3B + 3W)		Very low group (Grade 3BW)	
	No.	Per 1000	No.	Per 1000	No.	Per 1000	No.	Per 1000
Gross, uncorrected mortality	84	(15)	57	(37)	167	(43)	68	(93)
Anomalies	56	(10)	34	(22)	64	(17)	11	(15)
Premature birth	11	(2)	12	(8)	25	(6)	8	(11)
Net mortality corrected for anomalies and prematurity	17	(3)	11	(7)	78	(20)	49	(67)

*During the period 1960-1971: the totals in each diet group are the same as in Table III.

patients in my clinics. The mode was high protein and low carbohydrate with emphasis on red meat. The results are impressive and beyond any expectation. The difficulties in this program are well known to the author, but the continual demonstration of the benefits have only caused me to seek better ways of instruction of the pregnant patient and to improve further the basic system and the method of monitoring the progress of each woman.

In my view gestosis is at first a protective response to unfavorable conditions in pregnancy, specifically protein deficiency. With very poor diet this tendency becomes pathological, with grave clinical results for mother or child, sometimes both.

Improvement in protein intake, coupled with reduction in carbohydrate use, as monitored by an index that accounts for both weight gain and hemoglobin levels, results in a highly significant drop in the incidence of gestosis in a specific population. I might suggest that "normal" pregnancy be redefined to include only those parturients who achieve in the 38th week a hemoglobin above 12 g/dl (per deciliter or 100 ml) and a weight gain of less than 250 g/week. If such a standard were adopted I am confident that more sophisticated investigations would eluci-

date the physiological specifics of the cause of gestosis which I have solved in practice.

Likewise, such obstetric disasters as abruptio placentae, inexplicable fetal death, and the need for large numbers of abdominal deliveries due to inefficient uterus and birth canal will be brought down to extremely low frequencies.

I have neither the time nor resources to analyze fully the great mass of information I have collected over these many years. The overall rates of gestosis, fetal death, abruptio placentae, unresponsive labors, and other matters presented here speak to their own compelling conclusion. These methods can be equally applied elsewhere. With some form of national assistance a greater proportion of pregnant women can succeed against the economic bias of healthy pregnancy. Hospitalization during the antenatal period is used mainly for control of the diet. Young primigravidas are often admitted for 4 or 5 days after their first visit to the clinic. All who fail to keep to the diet at any maturity are admitted for instruction and control. Beds once occupied by various complications of pregnancy and labor are now used for intensive prophylaxis, and to a superior result.